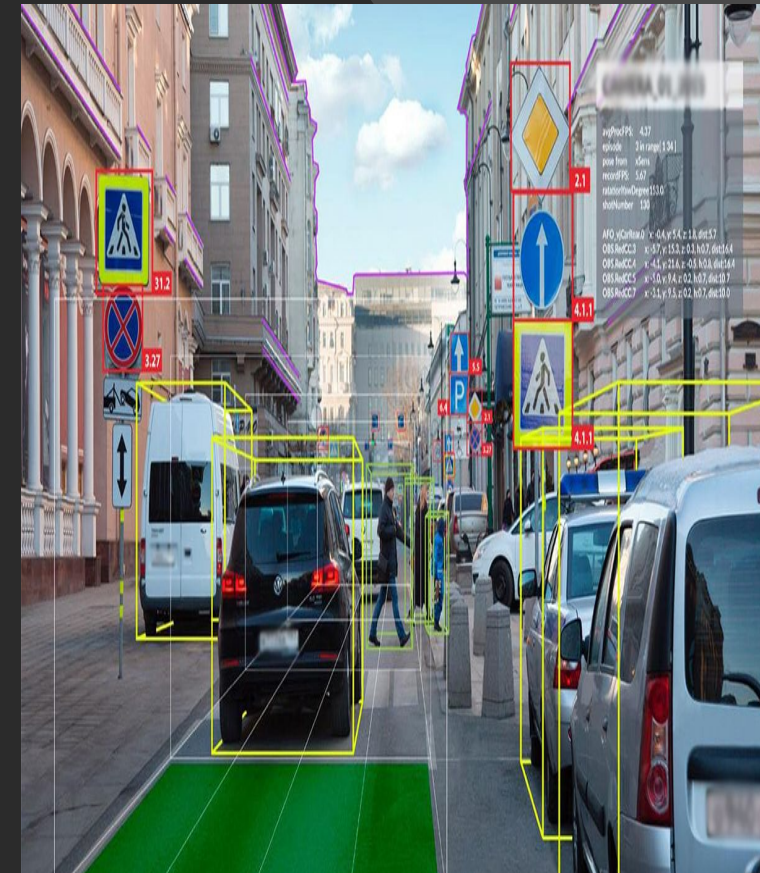




COMPUTER VISION IN AUTONOMOUS VEHICLES

Introduction:

- 1.Computer Vision helps a vehicle see and understand its surroundings using cameras and sensors.
- 2.It detects roads, pedestrians, vehicles, traffic signs, and traffic lights.
- 3.It processes images in real time to make safe driving decisions.
- 4.It is an important technology that helps self-driving cars drive safely without human control.



The Challenge: Seeing the World Like a Driver

An autonomous car must continuously detect roads, lane markings, pedestrians, vehicles, and traffic signs — then judge exactly how far away each object is. This requires three tightly connected building blocks of computer vision.



A. Vision Pipeline

The staged process that turns raw camera frames into meaningful detections: acquisition, preprocessing, feature extraction, detection, and decision-making.



B. Camera Calibration

Determining a camera's intrinsic and extrinsic parameters so that pixels can be accurately mapped to real-world coordinates.



C. Depth Estimation

Calculating how far away objects are using stereo vision, LiDAR fusion, or monocular deep-learning models — critical for safe navigation.

A. Vision Pipeline — From Pixels to Decisions



Each stage of the pipeline:

- Image Acquisition: Multiple cameras capture raw frames of the road scene
- Pre- processing: Noise removal, resizing, color correction, undistortion
- Feature Extraction: Edges, lane lines, textures and key points are identified
- Object Detection: CNNs classify pedestrians, vehicles, signs and lanes
- Decision & Control: Steering, braking and speed decisions are generated

B. Camera Calibration — Mapping Pixels to the Real World

Calibration Setup



Image Plane (u, v)

World Points (X, Y, Z)

Goal: find the mapping from 3-D world coordinates to 2-D pixel coordinates.

Intrinsic Parameters

- Focal length (f_x, f_y) — lens zoom factor
- Principal point (c_x, c_y) — optical center on sensor
- Lens distortion coefficients (radial & tangential)

Extrinsic Parameters

- Rotation matrix R — camera orientation on the car
- Translation vector t — camera position relative to the vehicle

Why it matters: accurate calibration lets the car convert what a camera "sees" into precise real-world distances and angles — essential before lane-keeping or obstacle-avoidance logic can run.

C. Depth Estimation — Judging Distance to Objects

Stereo Triangulation



baseline (b)

Left Cam

Right Cam

$$\text{Depth} = (\text{focal length} \times \text{baseline}) / \text{disparity}$$

Matching pixel disparity between two camera views reveals how far each point lies from the vehicle.



Stereo Vision

Two cameras compute disparity to triangulate distance geometrically.



LiDAR Fusion

Laser pulses measure exact distance; fused with camera images for accuracy.



Monocular Deep Learning

A single camera + neural network predicts a dense depth map directly.

Putting It Together: Real-Time Perception in Action



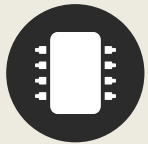
Lane and road-surface detection keeps the vehicle centered in its lane.



Pedestrian and cyclist detection triggers emergency braking when needed.



Traffic-sign and signal recognition informs speed and stopping decisions.



All three modules — pipeline, calibration, and depth — must run in real time on limited onboard hardware.



Conclusion

- A self-driving car's vision pipeline turns raw pixels into decisions;
- camera calibration guarantees those pixels correspond to accurate real-world geometry; and depth estimation tells.
- the car exactly how far away every road user is.
- Together, these three pillars of computer vision make safe, real-time autonomous driving possible.

